

Sparse Routing Improves Sample Efficiency in Small Language Models

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Abstract

Small language models are usually trained densely, which wastes capacity on tokens that need very little computation. We introduce sparse routing, a mechanism that sends each token to one of k lightweight expert blocks chosen by a learned router. On three benchmarks, sparse routing matches dense baselines with 41 percent fewer training tokens.

1 Introduction

Dense training treats every token as equally expensive. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

2 Related Work

Mixture-of-experts models have a long history. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

3 Method

Our method has three parts: a router, a set of experts, and a load term. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

3.1 The Router

The router is a single linear layer followed by a softmax over k experts. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

3.2 Load Balancing

Without a balancing term the router collapses onto one expert. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

4 Experiments

We evaluate on WikiText-103, C4 and a held-out code corpus. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

5 Results

Sparse routing reaches a perplexity of 18.4 on WikiText-103 against a dense baseline of 18.6, while using 41 percent fewer training tokens. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

6 Limitations

Our experiments are limited to models below one billion parameters. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

7 Conclusion

Sparse routing is a cheap way to buy sample efficiency. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension. We describe the construction in detail and analyse its behaviour under the assumptions stated above. The resulting procedure is straightforward to implement and requires no changes to the surrounding training pipeline. Throughout, we write n for the number of examples and d for the embedding dimension.

References

- [1] Shazeer et al. Outrageously large neural networks. ICLR 2017.
- [2] Fedus et al. Switch Transformers. JMLR 2022.